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Course Completed by

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on: May 06, 2026

Overview

In this course, learners explore how deep learning drives modern vision systems by building and understanding core models, including convolutional networks, transformers, and generative architectures such as diffusion models. They gain hands-on experience implementing and fine-tuning models for visual recognition and generation tasks, including image classification, object detection, segmentation, captioning, and image synthesis, while applying optimization techniques like gradient descent, batch normalization, and dropout. Learners also develop practical proficiency with modern deep learning frameworks and tools, enabling them to design, train, and evaluate large-scale vision models for real-world applications.

Competency Areas

Deep Learning Fundamentals

Convolutional Neural Networks

Attention and Transformers

Image Classification

Object Detection and Image Segmentation

Generative Models (GANs & VAEs & Diffusion)

Robot Learning and Deep Reinforcement Learning